

Play it safe or leave the comfort zone? Optimal content strategies for social media influencers on streaming video platforms

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ABSTRACT

Social media influencers (SMIs) endeavor to attract, retain, and, more importantly, influence large audiences through creating videos on streaming video platforms (SVPs). Facing diverse audiences and intense competition, SMIs who differ in their skills and performance levels struggle with two challenging trade-offs in video production decision: (1) trade-off between *reinforcement-seeking* (i.e., reinforcing a specialized content style to play it safe) and *variation-seeking* (i.e., varying content style to leave the comfort zone), and (2) trade-off between early and late strategic action. This study explores the sweet spot of reinforcement- and variation-seeking by proposing a finite-horizon Markov Decision Process (MDP) model to maximize SMIs' influence on audiences on SVPs. Leveraging a series of musical videos produced by SMIs, our findings show that low-performing SMIs benefit more from variation-seeking than from reinforcement-seeking. By contrast, the influence of high-performing SMIs on audiences could be maximized by reinforcement- or variation-seeking, depending on the SMI type. High-performing SMIs' variation-seeking (should it be taken) is encouraged earlier than later over a planning horizon. This study contributes on on-demand content decision making and makes recommendations to SMIs and SVPs.

1. Introduction

The prevailing streaming media and video-on-demand services have transformed the ways how videos are created and consumed [1] and fostered a growing streaming video market enabled by streaming video platforms (SVPs), e.g., YouTube and Vimeo [2]. For instance, YouTube hosts more than 500 h of video content uploaded every minute and attracts over 2 billion monthly active users [3]. According to the report published by Fortune Business Insights in 2022 [4], the global video streaming market is projected to grow from \$473.39 billion in 2022 to \$1690.35 billion by 2029. Reaching out to audiences through streaming videos, a new type of third-party endorsers, known as social media influencers (SMIs), are able to shape audiences' attitudes [5]. In contrast with text- and image-based media platforms, SVPs accommodate various digital stimuli (e.g., video, audio, and text), provide advantageous affordances (e.g., rich media formats, identity persistence, real-time feedback), and enable real-time interactions between content creators and viewers (e.g., live-streaming services) [6,7]. Making use of

such a dynamic and interactive way of streaming video production, SMIs rise to fame that could be translated into market value in the forms of advertising and sponsorship in online attention economy.

In contrast to well-known celebrities in the traditional mass media, SMIs are bottom-up grassroots who have gained fame as content generators on social media platforms [8]. The open access to SVPs allows people from all walks of life to create videos of varying quality. With the growing popularity of widely spread videos on SVPs, video creators as self-made SMIs rising to stardom are able to influence audiences, such as steering audiences toward voting [9] and stimulating audiences' gifts or tips [10,11]. In this study, we investigate SMIs' influence with a focus on online audiences' votes for their favorite SMIs.

SMIs could amass their influence on audiences through producing audience-facing contents that accommodate a wide range of audiences' preferences on SVPs. While SVPs feature videos in a variety of domains ranging from talking, eating, gaming, dancing to singing [12], SMIs may choose one specific domain as their signature domain. Even though a SMI focuses on a particular domain (e.g., singing), he or she still has the

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autonomy to produce various contents (e.g., different music genres), to not only please followers but also express themselves [13].

With multiple audiences collapsed into SVPs [14], SMI's videos are consumed by audiences exhibiting both consistency-seeking and variety-seeking tendencies [15], which have implications for SMIs' sequential decisions about what to produce and how to strategize video production. Scholars investigating consumers' content choices have identified mixed effects of content producers' repetitive offerings, including a positive effect (e.g., loyalty) encouraging consumers' consistency-seeking [16] and a negative effect (e.g., boredom) triggering consumers' variety-seeking [17]. Meanwhile, practitioners in the SMI industry have raised the concern about striking a balance between repetition and variety of content, e.g., decision making between high-volume-but-repetitive and thin-but-optimized content [18]. The viewpoints in both academia and industry suggest that there is no one-fits-all strategy for SMIs' content production on SVPs.

Being aware of audiences' hybrid consumption tendencies of consistency-seeking and variety-seeking, SMIs who have picked up their signature domain of video production are confronted with a trade-off between reinforcing and varying their content styles. Reinforcing the content style that a SMI is inherently specialized in to play it safe, i.e., *reinforcement-seeking*, renders content production consonant with SMI's innate nature. Although reinforcement-seeking is conducive to establishing SMI's core competitiveness, it might be too safe to extend influence on audiences. In contrast to reinforcement-seeking, varying the content style beyond SMI's expertise to leave the comfort zone, i.e., *variation-seeking*, could refresh and diversify SMI's content repertoire. Displaying a multifaceted image can prevent audiences from getting bored of repeated content style, yet it may incur risk due to uncertainty in audiences' feedback on new attempts and even dilute the salience of SMI's core style. Concerning such a trade-off between reinforcement and variation-seeking, our study aims to locate the sweet spot under which SMIs' videos yield maximum influence on audiences.

To turn one-off hits into winning streaks, SMIs strive to accumulate their influence on audiences through a multi-step content production process during a planning horizon on SVPs. Apart from the decision between reinforcement- and variation-seeking, SMIs need to decide when the chosen action should be taken, either early or late in the planning horizon, raising another trade-off concerning the timing of strategic actions. While seeking variation in early stage gives SMIs room for exploration, it makes difficult for them to establish a salient and consistent identity. Seeking variation in late stage, however, may weaken previously established image and even trigger audiences' resistance. Thus, the trade-off between early and late action of content strategy motivates us to identify the optimal timing. Considering the heterogeneity of SMI type, our study addresses the following three research questions.

1. Under which circumstances are reinforcement- and variation-seeking strategies effective for SMIs to maximize influence over audiences on SVPs?
2. How does the planning horizon affect the optimal timing of decision making between reinforcement and variation-seeking on SVPs?
3. How do the optimal reinforcement- and variation-seeking strategies vary across different types of SMIs on SVPs?

The extant literature has identified a set of factors for strategizing digital content production, such as content credibility [19] and textual characteristics appealing to the sentiment of the audience [20], video and audio characteristics of scenes arousing audience's emotion [21], characteristics of content creators [22,23], and active management of profiles in social networking sites [24,25]. However, several key issues remain unaddressed in prior research. First, although previous studies have extensively investigated how to attract and retain audiences [20,26,27], content strategies for influencing audiences to arouse favorable attitudes and behaviors remain unclear. Second, with multiple

audiences' mixed consumption tendencies (i.e., consistency- and variety-seeking), we have little knowledge about how SMIs reconcile the trade-off between reinforcement- and variation-seeking in content production. Third, although content strategies has been widely examined in user-generated content (UGC) platforms [20,27] and social media sites [28], how the highly dynamic and interactive SVPs contribute to SMIs' content strategies awaits further investigation.

Bearing the gaps in mind, we develop a Markov Decision Process (MDP) model [29] to identify the sweet spot of reinforcement- and variation-seeking strategies. In line with SMIs' forward-looking tendency [27,30], our MDP approach captures the inter-temporal dynamics between SMI's content strategies and their influence on audiences. To calibrate the parameters of the proposed MDP model, we collected and coded six weeks of SMIs' musical performance videos in a singing talent competition streamed from June 22 to August 3 in 2019 on a major SVP in China. The contestants of the competition went through eliminations based on audiences' votes on a weekly basis, and those who survived the week-by-week eliminations rose to the top as SMIs. Different types of SMIs and their reinforcements and variations in musical performance videos were identified via clustering analysis and their influence on audiences was measured by online audiences' weekly votes. As the musical performances were recorded by the producer of the competition program and streamed on the same SVP, the quality of the video content hinges on SMI's performance. Leveraging the dynamics between SMIs' video production and influence on online audiences, we estimated the MDP model's state transition probability and reward matrices using matrix completion techniques [31]. The MDP solution served as the optimal strategies for how and when to reinforce and/or vary video contents to obtain maximum influence through a multi-step process within a planning horizon.

Three key findings emerge from our computational model. *First*, the variation-seeking strategy favors low-performing SMIs in the sense that SMIs producing relatively low-quality videos are encouraged to seek variation in content style. *Second*, for high-performing SMIs, the effectiveness of variation-seeking strategy is highly state-dependent and differentially benefits certain types of SMIs with a likely decline in content quality. *Third*, for SMIs seeking variation in content style, the optimal timing of variation-seeking depends on the remaining time (to completion) over the planning horizon, advocating early rather than late variation.

Our study makes several contributions to the research on digital content strategy. *First*, this paper reveals the varying effectiveness of reinforcement- and variation-seeking strategies on SMI's influence on audiences, a more acute indicator of SMI's market value than attracting and retaining audiences. *Second*, the MDP model uncovers the highly heterogeneous effectiveness of opposing content strategies and are generalizable to inter-temporal decision making (e.g., budget allocation, video content pricing). *Third*, we add nuance to the existing streaming media studies by investigating the leveraging power of on-demand streaming media in enabling video creators to achieve SMI status. Our findings also provide important practical implications for SMIs and SVPs to derive value from strategic content production.

2. Literature review

2.1. Influencers' strategies of social media production

Recognizing the business value of social media, previous studies have investigated UGC strategies to compete for attention in a variety of social media contexts. In blogging sites, textual characteristics such as content sentiment, quality [20] play different roles in attracting and retaining audiences. Apart from designing social media content deliberately, differentiating content from peers is also a widely adopted strategy. For example, on online review platforms, top reviewers tend to avoid crowded review segments and post more differentiated ratings strategically in response to competition effect [32]. In photo sharing

communities, users' contents evolve as they form social ties, attempting to upload unique content to distinguish themselves from peers with similarity popularity status [33]. Typical content production strategies in social networking sites (SNSs) include tailoring user profiles to the strongest audience [24], emotional expression and person-related content in status updates based on audience size and diversity [34], and strategically sharing self-enhancing and self-verifying content depending on perceived audience as close and distance friends [25].

In sum, content strategies revolve around content formats (e.g., text, image, audio, and video), content characteristics (e.g., genre, quality, sentiment, and tone), and creator characteristics (e.g., popularity and status) [19,23,35]. Regarding the various content and creator characteristics in the nomological framework for explaining and predicting audience responses, Han et al. [35] emphasized the effect of interactions between content and creator characteristics on content virality (i.e., large-scale diffusion of social media content). Strategic content production on social media platforms enables creators to pursue an array of objectives, such as constructing digital identities [26], attracting and retaining audience [20,27], engaging audience with content [36], enhancing status [37], and monetizing content [38]. Regarding the various purposes that social media content serves, however, exerting influence on audiences, which is the primary goal for SMIs, have received little research attention. As social media platforms enable SMIs to earn distinction and status, SMIs are able to leverage strategic content production to influence audiences' attitudes and behaviors [37], which awaits in-depth investigations.

2.2. Audiences' choices of social media consumption

With the control, choice, mobility, and manipulability of digital content [39], audiences' choices of digital content consumption have their roots in consumers' choice behaviors, which suggest that consumer's consistency- and variety-seeking tendencies may coexist within consumer over time depending on his or her choice history [17,40,41]. The satisfaction obtained from consuming content results in people's consistency-seeking to routinize and reinforce satisfied consumption behavior [40]. When consumers get satiated or bored with familiar content, variety-seeking comes into play so that they seek out stimuli that are less familiar [41]. In the context of social media consumption, due to vastly divergent tastes, audiences respond differently to social media content [8], exhibiting reinforced and varied consumption patterns [15].

Apart from the within-consumer hybrid tendencies of consistency- and variety-seeking, different consumers' diverse, spontaneous, and unpredictable consumption choices of social media [42] result in heterogeneity in the degree of consistency- and variety-seeking tendencies across consumers. Audiences exhibit the choices of consuming multiple items from a diverse set of categories. Prior research on consumption of streaming music [43], for instance, found that audiences tend to increase the quantity and diversity of consumption and discover new content (e.g., new artists, songs, and genres). The networked audiences on social media platforms create new tensions when SIMs tend to combine public-facing content production and self-serving content distinction [44], spurring SMIs to adjust content production with tactical acumen. In the face of audience's fast-changing needs, SMIs are concerned with how and when to cater for audience's mixed consistency-seeking and variety-seeking tendencies.

2.3. Dynamics on streaming video platforms

In contrast to text- and image-based social media platforms, video-based social media platforms (e.g., YouTube) provide enriched multimedia affordances facilitating the production of videos and interaction with audiences [45], giving rise to high-profile SMIs with a large volume of video views [46]. Streaming video, as a new form of video service on social media platforms [47], reduces psychological distance between

content creators and audiences and cultivates co-experience in terms of participation, cognitive communion, and resonant contagion [42,48]. Unlike ownership-based consumption such as physical DVD or downloading, the access-based consumption enabled by streaming video services facilitates a sense of psychological ownership that results in audience's loyalty, empowerment, and social rewards [39].

SVPs are subject to the simultaneous presence of multiple audiences that make it difficult for SMIs to reconcile discrepancies among audiences under context collapse [34]. The social interactions between video creators and multiple audiences render the production and consumption of streaming video highly dynamic and contagious [46]. In particular, SMIs' self-expressing videos influence audiences through forging intimate, emotion-laden interpersonal connections with audiences [49], which differs from commercial UGC persuading audiences to purchase products or services [50,51].

Regarding SMIs' various purposes of content production (e.g., audience-influencing and self-expressing), prior research further investigated the factors affecting content creators' reinforcement- and variation-seeking strategies, such as content creators' intrinsic characteristics (e.g., demographic and personality traits) for reinforcement-seeking and content consumers' features (e.g., niche audience's attention) for variation-seeking [52]. While both reinforcement- and variation-seeking strategies have significant effects on content production and consumption, the dominance of the effect depends on the content type, e.g., music versus tech videos [53]. With respect to online music videos, the effect of music videos on music sales varies with creator features such as newcomer versus incumbent artist and content genre such as mainstream versus niche music genre [38].

Since the "rules" of connecting streaming videos with massive audiences remain ill-defined [54], SMIs struggle with balancing reinforcement- and variation-seeking strategies for streaming video production in a dynamic and strategic manner. However, the content strategies identified in prior studies across social media platforms revolve around static content and creator characteristics (see the summary in Table 1). The effectiveness of content strategies investigated in the literature was predominately analyzed by regression models, which hardly capture the inter-temporal nature of content strategies on SVPs. To address this research gap, our study adopts the MDP approach to modeling both content and creator characteristics and examine the dynamics between influencers and audiences.

3. MDP modeling inter-temporal influencer-audience dynamics

The influencer-audience dynamics motivate us to develop a MDP framework to capture the effectiveness of inter-temporal decision making on content production. The Markovian property is well aligned with the short active life span of videos on SVPs. With the video feeds on SVPs constantly exposing audiences with new content, the influence of historic videos decays dramatically and the impact of newly created videos on audiences is short-lived. Audiences' responses to SMIs' videos primarily depend on the latest content, which is in accord with the memoryless nature of MDP model. The MDP model also allows us to fully capture the dynamics between SMI's reinforcement- and variation-seeking actions and their influence on audiences.

Previous studies have extensively used Markov models to examine user dynamics across various contexts, for instance, gamers' latent engagement state [55], customers' engagement and satisfaction with a firm [56], and customer purchase motivation in online shopping [57]. When making decisions for content production in social media, SMIs tend to look forward, i.e., taking into account both the current-period and future benefits [27,30]. With SMIs' forward-looking tendency, we model the inter-temporal influencer-audience dynamics using a finite-horizon MDP model [29].

Drawing from the state-dependent optimal policy, we investigate the effectiveness of reinforcement- and variation-seeking on influence optimization for each SMI type. To calibrate the parameters of the MDP

Table 1

Prior studies on strategies for content production on social media platforms.

Illustrative research	Research context	Content format	Factors in content strategy	Research method
Huang, et al. (2015)	Blogging platform	Text	- Content sentiment - Content quality	Dynamic modeling
Zeng and Wei (2013)	Flickr	Image	- Social ties of creator dyads - Creator popularity - Content similarity	Linear regression analysis
Zheng, et al. (2020)	Facebook	Text and image	- Creator-audience ties - Content type (self-enhancing vs. self-verifying)	Experiment on Amazon Mechanical Turk
Han, et al. (2020)	Twitter	Text, image, and video	- Content characteristics (valence, topics) - Creator characteristics (originality, topic propensity)	Logistic regression analysis
Kretschmer and Peukert (2020)	YouTube	Music video	- Creator type (newcomer vs. incumbent) - Content type (mainstream vs niche)	Natural experiment
Qiu, et al. (2015)	YouTube	Video in general	- Content quality - Content diffusion patterns (learning vs. network effects) - Content type (music vs. tech)	Linear regression analysis
Zhao, et al. (2021)	Twitch	Streaming video in gaming	- Content characteristics (valence, topics) - Creator characteristics (personality, professionalism) - Streaming affordance	Linear regression analysis
Current Study	Tencent Video	Streaming video in musical performance	- Content decision (reinforcement vs. variation) - Creator type (low- vs. high-performing) - Decision timing (early vs. late)	MDP modeling

model (i.e., state transition matrix and reward function), we employ matrix completion approach to estimate the influencer-audience dynamics observed in real world. Previous studies [58,59] have provided theoretical underpinning for the claim that the true underlying matrix can be recovered with high accuracy. Multiple matrix completion methods (see Supplementary Appendix A) are used to triangulate the recovered matrices for state transition probabilities and reward function.

3.1. Empirical data collection and classification of states

3.1.1. Empirical setting for data collection

We collected the empirical data from a singing talent competition aired in 2019 on Tencent Video, a major SVP in China. The competition consisted of a series of musical performances produced by contestants, each of which performed one song on a weekly basis. Contestants made their public debut in the first round of the competition as audition that was evaluated by six expert judges to determine whether contestants

were able to continue participating in the competition. To survive the audition round, contestants made an all-out effort to perform their signature style. The performances in the remaining rounds were evaluated in turn by either six expert judges or a 300-person live studio audience, which was determined by the rules of the competition program. As contestants' performances were streamed on Tencent Video platform, online audience voted for their favorite contestants through social media platforms in each week. Since the second round of the competition, the total votes each contestant obtained within one week were counted and released week by week. Contestants at the bottom of the vote ranking were eliminated, and those who survived the competition earned the SMI status.

The contestants who made the top 14 list with massive votes received from online audience were selected as SMIs for our empirical study. These SMIs performed songs in various genres (such as pop, folk, rock, jazz, etc.), each of which lasted for three to five minutes, from June 22 to August 3 in 2019 on a weekly basis. For each round of the competition, we collected SMIs' musical performance video clips,¹ and each performance's ratings from expert judges and live studio audience. Online audience's casting their votes in the leaderboard of the contestants is an active behavior influenced by SMI's content. Through the action of voting for a specific SMI, a wide online audience expressed their positive attitude toward and support for the SMI. Audience votes thus become a manifestation of SMI's influence on audiences in the sense that online audience changed their stand from a passive viewer to an active advocate. To capture SMIs' influence on online audience, we collected the weekly counts of online votes and calculated each SMI's vote share, i.e., the percentage of online votes each SMI received in each week, as the immediate reward in the MDP model. In total, our dataset consists of 14 SMIs producing 86 performances across six weeks. The descriptive statistics of the online votes are present in Tables 2 and 3 respectively.

3.1.2. Coding of video content

Concerning the decisive role of online audience's votes in SMI's progression in the competition, content style of SMIs' musical performance videos was coded from the audience's perspective, i.e., ordinary viewers' subjective perceptions of SMIs' musical performances, rather than the objective acoustic features of videos. All video clips were coded by three independent raters with respect to three aspects, song-level genre [60], musically evoked emotion [61], and musical performer trait [62], including both content and creator characteristics [35]. Each aspect consists of multiple categorical labels, resulting in a multidimensional vector with 27 binary features used to characterize each performance video (see Supplementary Appendix B). The performance features were finalized by the consensus of the majority of the three coders' results. To ensure the inter-rater reliability of the coding results, we tested inter-rater agreement across the three coders using generalized Cohen's kappa coefficient for multivariate labels [63]. The kappa coefficients of the final coding results compared with that of the three raters respectively are 0.721, 0.707, and 0.683, suggesting the agreement in the substantial range [64].

3.1.3. Classification of states

SMIs' performance videos were recorded by the program producer and streamed on the same SVP, which controlled the potential confounding effect of video/audio quality on the content quality of musical performances. The states of our MDP model capture two key features of SMIs' content production, i.e., content quality and content style, both of which are explicitly observable. The evaluations of expert judges and live studio audience approximate to content quality from the expert's point of view, whereas the multidimensional features derived from the video coding reflect content style from ordinary audience's perspective.

¹ An example of the video clips can be found at <https://www.youtube.com/watch?v=KpfmKvGbePg>.

Table 2

Descriptive statistics of audience votes for each round of performance.

Round	Weekly count of online votes					Weekly share of online votes			
	Total	Mean	S.D.	Min	Max	Mean	S.D.	Min	Max
1	10,568,656	754,904	767,947	105,906	2,482,935	7%	7%	1%	23%
2	26,099,648	1,864,261	1,458,457	207,689	4,194,452	7%	6%	1%	16%
3	57,921,878	4,137,277	2,083,816	1,232,064	6,638,868	7%	4%	2%	11%
4	81,993,252	5,856,661	2,616,139	2,182,664	9,070,094	7%	3%	3%	11%
5	104,553,919	7,468,137	3,146,167	3,039,509	11,678,306	7%	3%	3%	11%
6	130,776,399	9,341,171	4,046,148	3,394,876	15,967,582	7%	3%	3%	12%

Table 3

Descriptive statistics of online votes for each SMI.

ID	Weekly count of online votes					Weekly share of online votes				
	Total	Mean	S.D.	Min	Max	Sum	Mean	S.D.	Min	Max
1	48,366,995	8,061,166	4,597,499	2,482,935	14,810,742	84%	14%	5%	11%	23%
2	25,935,530	4,322,588	2,736,516	634,069	7,648,205	38%	6%	1%	6%	7%
3	42,087,376	7,014,563	4,133,600	1,690,249	12,686,893	69%	11%	2%	10%	16%
4	36,766,347	6,127,725	3,999,401	595,234	10,927,652	52%	9%	2%	6%	10%
5	10,486,208	1,747,701	1,349,085	189,199	3,394,876	14%	2%	1%	1%	3%
6	12,085,811	2,014,302	1,799,630	200,461	4,855,225	15%	2%	1%	1%	4%
7	47,327,865	7,887,978	5,533,915	993,111	15,967,582	67%	11%	1%	9%	12%
8	26,707,420	4,451,237	3,115,226	184,064	8,066,311	34%	6%	2%	2%	7%
9	39,102,540	6,517,090	4,460,966	389,697	12,116,541	53%	9%	3%	4%	10%
10	21,290,588	3,548,431	2,866,637	265,692	7,477,402	26%	4%	2%	2%	6%
11	45,812,003	7,635,334	4,160,361	2,039,251	13,116,475	78%	13%	4%	10%	19%
12	10,345,740	1,724,290	1,412,088	105,906	3,448,836	12%	2%	1%	1%	3%
13	19,729,713	3,288,286	2,738,585	227,151	7,073,025	24%	4%	2%	2%	5%
14	25,869,616	4,311,603	3,442,606	571,637	9,186,634	34%	6%	1%	4%	7%

According to the evaluation rules of the competition, contestants' performances in the first (audition), second, fourth, and sixth rounds were evaluated by the six-person panel of expert judges, whereas the performances in the third and fifth rounds were evaluated by the 300-person live studio audience. Each person in the expert panel and live studio audience could contribute one score to the video content quality, resulting in different ranges of scores for different rounds of performances. To make the quality scores consistent across different rounds, we rescaled the quality scores rated by the live studio audience from the range between zero and 300 to the range of zero to six to match up with the range of expert judges' scores.

The content quality scores were then dichotomized into two levels. We conducted a hierarchical clustering analysis, which takes into account the variances of the data points. Pruning the dendrogram results into two clusters, we identified 36 videos of low quality (with mean of quality score at 3.39) and 48 videos of high quality (with mean of quality score at 5.40) given that multiple videos could have the same quality score. To test the robustness of the dichotomization, we also applied median splitting to the quality scores. The median score of five divided the videos into the same allocation of low (zero to four) and high (five to six) levels with the hierarchical analysis.

As content style of each video clip was labeled as a multidimensional vector with 27 features, we conducted a partitioning around medoids (PAM) clustering [65] with the number of clusters estimated by

optimum average silhouette width [66]. Hamming distance [67] was used for the PAM clustering given the binary coding for the features of video content. The clustering analysis suggests three clusters, i.e., rocking, expressive, and lovely, each of which represents a distinctive content style according to the features of the medoid in each cluster. Integrating the three content styles with two levels of content quality of musical performance videos, content characteristics consist of six states {1 = Rocking-Low, 2 = Expressive-Low, 3 = Lovely-Low, 4 = Rocking-High, 5 = Expressive-High, 6 = Lovely-High}, which are observable on a weekly basis as SMIs' performance videos are streamed for each round of the competition. Table 4 presents the descriptive statistics of the content quality score and vote share received by each state.

Given that debut performance in the first round (i.e., audition) features each SMI's signature performance, the content style of debut performance represents the content style each SMI is expert in and the content quality of debut performance suggests whether a SMI is low- or high-performing with respect to musical content production. According to the content style and quality of each SMI's debut performance, we identified different types of SMIs in terms of inherit content style (i.e., rocking, expressive, and lovely) and capability of content production (i.e., low- versus high-performing). While SMIs' innate characteristics play a key role in defining SMI types, they could switch across a wide spectrum of state space to seek variation. The state to which each SMI's first round of performance belongs is considered as their initial state. Starting

Table 4

Descriptive statistics of state space.

State	State description	Frequency	Percent	Content quality				Vote share (%)			
				Mean	S.D.	Min	Max	Mean	S.D.	Min	Max
1	Rocking-Low	12	14%	3.17	1.27	0	4	7%	3%	3%	12%
2	Expressive-Low	12	14%	3.67	0.49	3	4	5%	4%	1%	16%
3	Lovely-Low	12	14%	3.33	0.78	2	4	5%	3%	1%	10%
4	Rocking-High	16	19%	5.31	0.48	5	6	8%	5%	2%	23%
5	Expressive-High	27	32%	5.48	0.51	5	6	9%	5%	1%	19%
6	Rocking-Low	5	6%	5.20	0.45	5	6	5%	2%	2%	6%

with the initial state, SMIs could either reinforce their proficient style and quality by maintaining the same state or vary the content style and quality by experimenting a different state.

3.2. MDP model specification

3.2.1. Finite-horizon MDP flow

The proposed finite-horizon MDP model consists of five major components: (1) the set of states $\mathcal{S}$, (2) the set of actions $\mathcal{A}$; (3) the state transition matrix $\mathcal{P}$, (4) the rewards $\mathcal{R}$ received from a given state and an action, and (5) the initial state s_1 , all of which can be specified as a tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, S_1)$. Fig. 1 graphically illustrates how an SMI makes a decision of content production, transits between states, and receives reward based on state and action. The notations used throughout the paper are listed in Table 5.

Our finite-horizon MDP works as follows. The SMI and SVP interact in a sequence of discrete time steps, $t = 1, 2, \dots, T$. SMI starts from a state s_1 at round $t = 1$. For each time step t , SMI resides in a state $S_t \in \mathcal{S}$, and takes an action of reinforcement- or variation-seeking, $A_t \in \mathcal{A}$ for the production of streaming media content. The action taken at t determines whether she will either transit to a different state or remain in the current state. As a result of the action, she receives a numerical reward, $R_t \in \mathcal{R} \subset \mathbb{R}$, and probabilistically gets into the next state S_{t+1} . At the terminal time step, she finishes production with the accumulated rewards in return, thereby giving rise to a sequence below over the T time periods in units of weeks.

$$H = (S_1, A_1, R_1, S_2, A_2, R_2, \dots, S_T, A_T, R_T) \quad (1)$$

The sets of states and actions (i.e., $\mathcal{S}$, and $\mathcal{A}$) have a finite number of elements. We index $\mathcal{S}$ from 1 to N . Based on the empirical clustering results discussed in Section 4, we have $N = 6$ number of states, namely, $\mathcal{S} = \{1 = \text{Rocking-Low}, 2 = \text{Expressive-Low}, 3 = \text{Lovely-Low}, 4 = \text{Rocking-High}, 5 = \text{Expressive-High}, 6 = \text{Lovely-High}\}$. For a particular SMI, her SMI type is given by the initial state S_1 , representing her intrinsic style displayed at the start of the streaming season. Subsequently, the state sequence is $(S_1, S_2, \dots, S_T)$, where $S_t \in \mathcal{S} = \{1, 2, \dots, N\}$. The action sequence is $(A_1, A_2, \dots, A_T)$, where $A_t \in \{0, 1\}$ equals 0 if a reinforcement-seeking action is taken and equals to 1 if a

Table 5
Summary of notations.

Notation	Description
T	Length of finite-horizon in units of weeks
N	Maximum index of state
H	Sequence of state, action, reward throughout the finite-time horizon
R	Expected total reward throughout the finite-time horizon
t	Time index at week level, $t \in \{1, 2, \dots, T\}$
S_t	State at time t , $S_t \in \{1, 2, \dots, N\}$
A_t	Action taken at time t , $A_t \in \{0, 1\}$
R_t	Reward received at time t , $R_t \in \mathbb{R}$
$P(s' s, a)$	State transition probabilities matrix from state s to s' with the action a , $P(s' s, a = 0) = 1$ when SMI decides to reinforce the current state such that $s' = s$
$p(s, s')$	Probability of state transition for $P(s' s, a = 1)$ when SMI decides to seek variation such that $s' \neq s$
$r(s, a)$	Weighted average expected reward received for state s with the action a
$r(s, a, s')$	Expected reward received from state transition from state s to s' with the action a
$V_k^\pi(s)$	Expected accumulative reward obtained by following policy π with current state s at time k
$\pi_k^*(s)$	Optimal policy for state s at time k
$V_k^*(s)$	Expected accumulative reward obtained by following optimal policy with state s at time k

variation-seeking action is taken. Given the state-action pair (S_t, A_t) , the SMI receives a reward R_t in the form of voting share.

3.2.2. State transition matrix and reward function

With a current state s , an action of reinforcement-seeking ($a = 0$) makes the next state same as the current state. By contrast, an action of variation-seeking ($a = 1$) renders the next state s' uncertain. The possibility of transiting to a different state and the reward derived from the transition vary with the next state. The state variable S_t , which depends on the preceding state S_{t-1} and action A_{t-1} , follows the transition matrix $\mathcal{P}$, which is defined as a three-argument function $P(s'|s, a) : \mathcal{S} \times \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$ as follows:

$$P(s'|s, a) = \Pr\{S_t = s' | S_{t-1} = s, A_{t-1} = a\}, \quad (2)$$

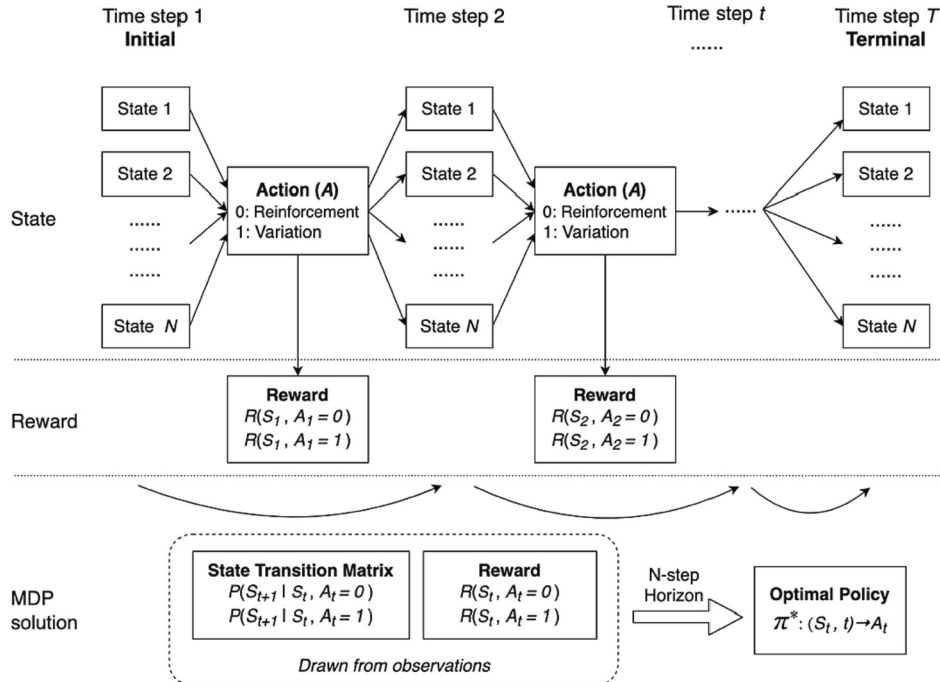


Fig. 1. Finite-horizon MDP flow chart.

where $\sum_{s' \in \mathcal{S}} P(s'|s, a) = 1$, for all $s \in \mathcal{S}$, $a \in \mathcal{A}$. Based on the assumption that SMI remains the same state if she chooses to reinforce the current state ($a = 0$) and decisively transitions to a different state if she chooses to seek variation ($a = 1$), we define $P(s'|s, a)$ as follows:

$$P(s'|s, a) = \begin{cases} 1, & \text{if } a = 0 \text{ and } s' = s \\ 0, & \text{if } a = 0 \text{ and } s' \neq s \\ 0, & \text{if } a = 1 \text{ and } s' = s \\ p(s, s'), & \text{if } a = 1 \text{ and } s' \neq s \end{cases}, \quad (3)$$

where

$$p(s, s') = \Pr\{S_t = s' | S_{t-1} = s, A_{t-1} = 1, s' \neq s\}. \quad (4)$$

With respect the reward function, a three-argument function $r(s, a, s')$, i.e., the expected reward based on the current state s , action a , as well as the next state s' , is used to define the reward for state-action-next-state triple $r(s, a, s') : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$ as follows:

$$r(s, a, s') = E[R_t | S_t = s, A_t = a, S_{t+1} = s'], \quad (5)$$

which refers to the reward when an agent at a state s takes an action a and arrives at a state s' in the next time step. We then define the reward expectation for state-action pair using a weighted average of all possible next states s' as follows:

$$r(s, a) = \sum_{s' \in \mathcal{S}} r(s, a, s') P(s'|s, a). \quad (6)$$

In the empirical setting, given a current state of Rocking-Low and an action of variation-seeking ($s = 1, a = 1$), SMIs transit to different next states at different frequencies, i.e., twice to Expressive-Low, once to Lovely-Low, once to Rocking-High, four times to Expressive-High, and once to Lovely-High, and the rewards observed for the transitions from Rocking-Low to other states range from 4.83 to 10.87. Taken together, we use the state-action-next-state triple to define and estimate the state transition $P(s'|s, a)$ and reward $r(s|s, a)$ functions to model the dynamics of SMIs' content production.

3.2.3. Dynamic programming for MDP

Based on the dynamic programming by the backward induction algorithm [68], the Bellman equation for the value function is further denoted as follows:

$$\begin{aligned} V_k^\pi(s) &= \left\{ r_k(s, a) + \sum_{s' \in \mathcal{S}} p_k(s'|s, a) V_{k+1}^\pi(s') \right\}_{a=\pi(s, k)} \\ &= \left\{ \sum_{s' \in \mathcal{S}} p_k(s'|s, a) r_k(s, a, s') + \sum_{s' \in \mathcal{S}} p_k(s'|s, a) V_{k+1}^\pi(s') \right\}_{a=\pi(s, k)} \\ &= \left\{ \sum_{s' \in \mathcal{S}} p_k(s'|s, a) [r_k(s, a, s') + V_{k+1}^\pi(s')] \right\}_{a=\pi(s, k)}, k = T-1, \dots, 1 \end{aligned} \quad (7)$$

which starts with $V_T^\pi(s) = r_T(s, \pi(s, T))$, i.e., the reward of terminal step given a state s and the action dictated by policy $\pi(s, T)$. Accordingly, the optimal value function is defined as follows:

$$\begin{aligned} V_k^*(s) &= \max_{\pi \in \Pi} V_k^\pi(s) = \max_{a \in \mathcal{A}} \left\{ \sum_{s' \in \mathcal{S}} p_k(s'|s, a) [r_k(s, a, s') + V_{k+1}^*(s')] \right\}, k \\ &= T-1, \dots, 1 \end{aligned} \quad (8)$$

where $V_T^*(s) = \max_{a \in \mathcal{A}} \{r_T(s, a)\} = \max\{r_T(s, 0), r_T(s, 1)\}$. Therefore, an optimal policy π^* satisfies

$$\pi_k^*(s) \in \arg \max_{a \in \mathcal{A}} \left\{ \sum_{s' \in \mathcal{S}} p_k(s'|s, a) [r_k(s, a, s') + V_{k+1}^*(s')] \right\}, k = T-1, \dots, 1 \quad (9)$$

and maximizes the expected total reward $V_1^*(s)$ given each initial state $s_1 = s$. The details about the state transition matrix, reward function, and accumulated reward are specified in Supplementary Appendix C.

4. MDP estimation results

According to the frequency of each state in Table 4, SMIs' performances cover all six states specified in the MDP model. As performances transit from state to state during the round-by-round competitions, the state transitions and associated rewards are represented by two matrices in the dimension of 6×6 . Through matching the observed state transitions and rewards with the 6×6 matrices, we identify seven unobserved elements out of the 36 elements, demonstrating dense matrices for the two MDP parameters. To estimate the unobserved elements, matrix completion techniques are applied to the state transition and reward matrices respectively. With the state and action spaces observed in the empirical setting, the state transition and reward matrices are estimated using five matrix completion methods, including simple mean, singular value decomposition (SVD), singular value thresholding (SVT), adaptive trace norm (ATN), and nonnegative matrix factorization (NMF) (see Supplementary Appendix C). In line with the five matrix completion methods implemented to recover $P(s'|s, a)$ and $R(s|s, a)$, the estimation results of the MDP model are reported for each method.

We first compute the expected reward (i.e., the weighted average reward of all possible outcomes) of each SMI state upon the reinforcement- and variation-seeking actions. While the reward for reinforcement-seeking adheres to $R(s|s, a=0)$ with certainty (i.e., $P(s'|s, a=0) = 1$), the expected reward for variation-seeking depends on the reward $R(s'|s, a=1)$ and the corresponding transition probability $P(s'|s, a=1)$. Given a current state s and the action of variation-seeking ($a=1$), the reward has multiple possible outcomes, each of which is specified with a possibility and reward value estimated in the state-action-next-state triple of transition probabilities $P(s'|s, a)$ and reward $R(s'|s, a)$. Table 6 presents the weighted average of state-action pair reward estimations, i.e., $\sum_{s' \in \mathcal{S}} R(s'|s, a) P(s'|s, a)$, for the five matrix completion methods.

The reward in our context represents SMI's influence on audience in the form of vote share on each state-action pair. The results in Table 6 show that reinforcement- and variation-seeking strategies result in different reward patterns across SMI states. Specifically, compared with reinforcement-seeking ($a=0$), the action of variation-seeking ($a=1$) produces higher reward for low-performing SMIs who have relatively low capability of musical performance. High-performing SMIs benefit from reinforcement- and variation-seeking differently. Whereas reinforcement-seeking outstrips variation-seeking for high-performing SMIs who are proficient in rocking or expressive style, the opposite holds for high-performing SMIs who are good at producing lovely content. Such reward patterns remain across the five matrix completion methods. In particular, the Lovely-Low state bears the lowest reward with reinforcement-seeking strategy ($a=0$), whereas it benefits most from variation-seeking strategy given the highest increase in reward.

We then examine the dynamics and degrees of variation-seeking by analyzing the characteristics of state transitions based on the transition probabilities. Considering that the state transitions occurring with high probability are the variations preferred by SMIs, we summarize each state's arrival states with the highest probability (see Table 7), which exhibit different patterns across SMI states.

Regarding the degree of variations engendered by state transitions, the degree of variation in content style and quality between two

Table 6

Expected reward estimations for state-action pair.

Action	Mean		SVD		SVT		ATN		NMF	
State	$a = 0$	$a = 1$	$a = 0$	$a = 1$	$a = 0$	$a = 1$	$a = 0$	$a = 1$	$a = 0$	$a = 1$
Rocking-Low (1)	5.47	7.80	5.47	7.80	5.47	7.80	5.47	7.80	5.47	7.80
Expressive-Low (2)	4.62	6.18	4.73	6.26	6.19	6.10	4.73	6.28	5.40	6.32
Lovely-Low (3)	2.09	5.86	2.09	5.86	2.09	5.85	2.09	5.86	2.09	5.86
Rocking-High (4)	11.33	7.13	11.33	7.13	11.32	7.13	11.33	7.13	11.33	7.13
Expressive-High (5)	9.54	7.79	9.54	7.80	9.54	7.79	9.54	8.06	9.54	7.75
Lovely-High (6)	4.83	6.77	4.02	6.76	5.96	6.17	4.40	6.37	5.09	6.75

Table 7

Variation-seeking dynamics based on MDP state transitions.

Current State		Next State with Highest Probability			State Transition Probabilities (SVD Method)					
		Next State	Degree of Variation		[,1]	[,2]	[,3]	[,4]	[,5]	[,6]
			Content Style	Content Quality						
Rocking-Low	[1,]	[5]	4	↑	0	0.222	0.111	0.111	0.444	0.111
Expressive-Low	[2,]	[4]	4	–	0.167	0	0.083	0.417	0.250	0.083
Lovely-Low	[3,]	[2]	3	–	0.125	0.375	0	0.250	0.125	0.125
Rocking-High	[4,]	[2]	4	↓	0.231	0.308	0.077	0	0.231	0.154
Expressive-High	[5,]	[1]	4	↓	0.333	0.083	0.25	0.25	0	0.083
Lovely-High	[6,]	[5]	3	–	0.182	0.182	0.182	0.091	0.364	0

Notes: The degree of variation in content style is measured by the Hamming distance index, which is widely used to measure the distance between two multivariate vectors with binary elements.

The degree of variation in content quality is denoted as ↑ if there is an increase in content quality from current state to next state, as – if there is no change in content quality from current state to next state, and as ↓ if there is a decrease in content quality from current state to next state.

consecutive states, which are measured by the pair-wise Hamming distance of content style and the change in content quality, also differs greatly across states. As shown in Table 7, the high-probability transitions suggest different degrees of variations in content style. Regarding variations in content quality, it is worth noting that certain types of high-performing SMIs are subject to a decrease in content quality when they

transit from their specialized style to a different style. Such patterns on variation-seeking dynamics are consistent across matrix completion methods.

To obtain the optimal policies, we solve the finite-horizon MDP model using dynamic programming with backward induction. As we leverage multiple matrix completion methods for state transition

Table 8

MDP optimal policies by matrix completion methods.

Matrix Completion Method	Content State	Action upon Remaining Time Horizon (T)											
		12	11	10	9	8	7	6	5	4	3	2	1
Mean	Rocking-Low	1	1	1	1	1	1	1	1	1	1	1	1
	Expressive-Low	1	1	1	1	1	1	1	1	1	1	1	1
	Lovely-Low	1	1	1	1	1	1	1	1	1	1	1	1
	Rocking-High	0	0	0	0	0	0	0	0	0	0	0	0
	Expressive-High	1	1	1	1	1	0	0	0	0	0	0	0
	Lovely-High	1	1	1	1	1	1	1	1	1	1	1	1
SVD	Rocking-Low	1	1	1	1	1	1	1	1	1	1	1	1
	Expressive-Low	1	1	1	1	1	1	1	1	1	1	1	1
	Lovely-Low	1	1	1	1	1	1	1	1	1	1	1	1
	Rocking-High	0	0	0	0	0	0	0	0	0	0	0	0
	Expressive-High	1	1	1	1	1	0	0	0	0	0	0	0
	Lovely-High	1	1	1	1	1	1	1	1	1	1	1	1
SVT	Rocking-Low	1	1	1	1	1	1	1	1	1	1	1	1
	Expressive-Low	1	1	1	1	1	1	1	1	1	1	1	0
	Lovely-Low	1	1	1	1	1	1	1	1	1	1	1	1
	Rocking-High	0	0	0	0	0	0	0	0	0	0	0	0
	Expressive-High	1	1	1	1	0	0	0	0	0	0	0	0
	Lovely-High	1	1	1	1	1	1	1	1	1	1	1	1
ANT	Rocking-Low	1	1	1	1	1	1	1	1	1	1	1	1
	Expressive-Low	1	1	1	1	1	1	1	1	1	1	1	1
	Lovely-Low	1	1	1	1	1	1	1	1	1	1	1	1
	Rocking-High	0	0	0	0	0	0	0	0	0	0	0	0
	Expressive-High	1	1	1	1	0	0	0	0	0	0	0	0
	Lovely-High	1	1	1	1	1	1	1	1	1	1	1	1
NMF	Rocking-Low	1	1	1	1	1	1	1	1	1	1	1	1
	Expressive-Low	1	1	1	1	1	1	1	1	1	1	1	1
	Lovely-Low	1	1	1	1	1	1	1	1	1	1	1	1
	Rocking-High	0	0	0	0	0	0	0	0	0	0	0	0
	Expressive-High	1	1	1	1	1	0	0	0	0	0	0	0
	Lovely-High	1	1	1	1	1	1	1	1	1	1	1	1

probabilities and reward functions, the results of optimal policies are computed for each method accordingly. With the maximum length of time horizon at 12 (the typical length of a streaming season), Table 8 presents the results of optimal policies, which are consistent across multiple matrix completion methods. Low-performing SMIs are recommended to seek variation in content constantly throughout the entire time horizon.

By contrast, the optimal policies for high-performing SMIs depend on the content state and remaining time horizon. Specifically, the Rocking-High state is recommended to reinforce the content state throughout the time horizon, while the Lovely-High state is recommended to vary the content state all the time. The Expressive-High state exhibits the nature and necessity of time-varying policy as the remaining time steps increase. With limited time horizon left, the optimal policy advocates reinforcement-seeking strategy ($\alpha = 0$) for Expressive-High state (see Table 8). When the length of time horizon suffices, the optimal policy favors variation-seeking ($\alpha = 1$) during the early steps, and switches to reinforcement-seeking ($\alpha = 0$) in the late phase.

We further compare the MDP model estimations with empirical observations in terms of inter-temporal dynamics and accumulated rewards, given the observed horizon sequences with $T = 5$ starting with five initial states (i.e., $s_1 \in \{1, 2, 3, 4, 5\}$). While each initial state ($s_1 \in \mathcal{S}$) represents a distinct SMI type endorsed by individual capability, an SMI featuring a specific type can demonstrate multiple content states that divert from her intrinsic type. Fig. 2 shows the action dynamics of observed five-step horizon in terms of degree of variation-seeking in content style indicated by average Hamming distance of two consecutive content styles for each initial state (i.e., SMI type). As opposed to the MDP optimal policies suggesting different actions to different SMI types, variation-seeking ($\alpha = 1$) is observed as the dominant policy across SMI types throughout the time horizon.

With respect to total rewards accumulated through the time horizon, we compare the expected rewards obtained by following the optimal policies with the observed rewards in the empirical setting. Starting with the same initial state, the same type of SMIs could take different courses of actions, thereby producing various reward sequences. Table 9 presents the observed rewards averaged by initial state and the observed policies executed by SMIs.

The expected rewards estimated by the MDP model and their differences with observed rewards are summarized in Table 10 according to the results of SVD method (see the comparison results of all five matrix completion methods in Supplementary Appendix D). Given the five initial states observed in the empirical setting, following optimal policies could yield higher expected rewards than conforming to SMIs' own actions observed in our empirical data. For the Lovely-Low type, the expected total reward following the optimal policy even rises to four

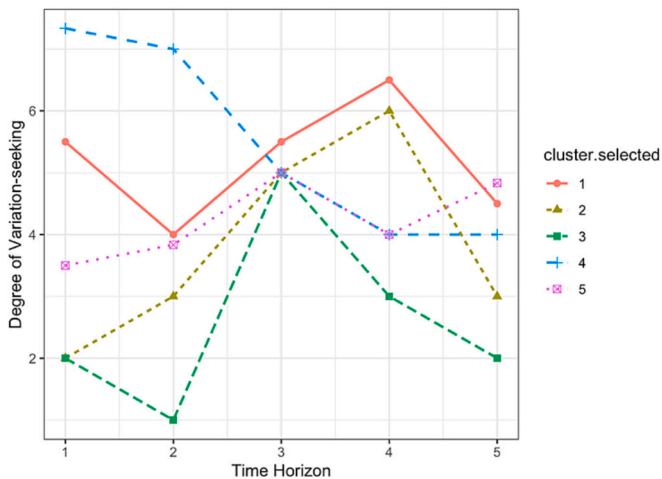


Fig. 2. Variation-seeking dynamics observed in the five-step horizon.

times of its observed total reward (i.e., 44 versus 11.85).

Apart from the comparison of optimal policies with the observed actions taken by SMIs, we further conducted counterfactual analysis by comparing optimal policies with commonsensical strategies to evaluate the effectiveness of the content strategies suggested by optimal policies. Specifically, we included four counterfactual policies as benchmarks, i.e., reinforcement all the time, variation all the time, early variation, and late variation, to estimate the rewards for the five-step horizon using the five matrix completion techniques respectively. The results of expected rewards for the four counterfactual policies with the MDP state transition and reward matrices (see Supplementary Appendix E1-E5 for details) underperform the optimal policies in Table 8. The results exhibit the same patterns across matrix completion technique.

5. Discussion and conclusion

In our empirical setting, although the contestants who survived more rounds of eliminations and made the top list are more likely to be qualified as SMIs, the number of eliminated contestants was determined by the competition rules. With top SMIs advancing to later rounds, the early rounds witnessed more contestants' performances. The consistent results of complete state transition and reward matrices recovered by different matrix completion methods suggest that the sample of observed videos is able to produce reliable results. However, it is noteworthy that the unobserved state transitions and rewards bring about marginal variations across matrix completion methods in the expected reward estimations in Table 6 and MDP optimal policies in Table 8. To test the robustness of the findings, we reconstructed the data set by including additional contestants who made the top list in an earlier round to build the MDP model. We conducted the robustness analysis by tuning the number of SMIs to four levels of top list (see Supplementary Appendix F). The consistent results of MDP model across the different date sets suggest the robustness of our findings.

The results of MDP estimations show that reinforcement- and variation-seeking strategies benefit low- and high-performing SMIs differently, uncovering three key findings on the effectiveness and timing of the two strategies for SMIs to achieve optimal influence on audiences on SVPs. First, variation-seeking (i.e., leaving the comfort zone) outweighs reinforcement-seeking (i.e., playing it safe) for low-performing SMIs on SVPs. Varying content style is more effective than reinforcing the same content style for low-performing SMIs to exert influence on audiences. Second, for high-performing SMIs, the effectiveness of reinforcement-seeking and variation-seeking depends on the SMI type. Reinforcement-seeking is recommended to certain type of high-performing SMIs (e.g., SMIs specializing in rocking music in our context), whereas variation-seeking is recommended to other type (e.g., SMIs with proficiency in lovely music). The degree of variation-seeking exhibits heterogeneity among different SMIs. High-performing SMI's variation in content style compensates for a likely decline in content quality. Third, for certain type of high-performing SMIs (e.g., SMIs featuring expressive musical style), the recommendation of reinforcement-seeking and variation-seeking is sensitive to the remaining time horizon in the sense that the optimal policy advocates variation-seeking (should it be taken) at early time steps and reinforcement-seeking at late stage. This is probably because variation-seeking generates a higher long-term reward when the remaining time is sufficiently long. If the remaining time is shorter than a certain threshold, reinforcing the status quo is superior to taking a variation-seeking action.

While the empirical context is a structured singing competition program, the concepts involved in the findings (i.e., content quality, reinforcement- versus variation-seeking strategies, and early versus late decision) and their operationalizations are generalizable to content production across different contexts. Although the competition rules determine the frequency of SMIs' content production and audiences' voting, how the content quality, content strategies, and timing of

Table 9
Empirical observations on rewards and policies.

Initial State (SMI Type)	Observed Reward					Observed Policy				
	Remaining Time Horizon ($T = 5$)					Remaining Time Horizon ($T = 5$)				
	5	4	3	2	1	5	4	3	2	1
Rocking-Low	30.50	25.43	19.29	12.92	6.44	1	1	1	1	1
Expressive-Low	39.50	31.78	23.31	15.10	7.26	1	0/1	1	1	1
Lovely-Low	11.85	10.41	8.17	5.51	2.60	0	0	1	1	0
Rocking-High	35.24	28.61	21.72	14.68	7.48	1	1	1	0/1	0/1
Expressive-High	40.40	31.56	23.58	15.73	7.93	0/1	0/1	1	0/1	0/1

Table 10
Comparison between MDP estimations and empirical observations (SVD method).

MDP Estimation						Difference with Observed Reward				
Initial State	Expected Reward					Difference Percentage				
	Remaining Time Horizon ($T = 5$)					Remaining Time Horizon ($T = 5$)				
	5	4	3	2	1	5	4	3	2	1
Rocking-Low	45.26	35.29	25.51	16.09	7.80	48%	39%	32%	25%	21%
Expressive-Low	46.19	35.81	25.56	15.71	6.26	17%	13%	10%	4%	-14%
Lovely-Low	44.03	33.72	23.67	14.05	5.86	271%	224%	190%	155%	126%
Rocking-High	56.65	45.32	33.99	22.66	11.33	61%	58%	56%	54%	51%
Expressive-High	47.70	38.16	28.62	19.08	9.54	18%	21%	21%	21%	20%

decision making are measured is transferable to other contexts of video production on SVPs. The audiences' voting for SMIs as the signal of SMIs' influence on audience is widely observed as a typical influencer-audience interaction. The way videos are produced and streamed by the program producer is conducive to controlling the technical aspects of video making.

This study makes three important theoretical contributions. First, the paper uncovers the varying effects of SMIs' reinforcement- and variation-seeking strategies on influencing audiences in terms of arousing their favorable behaviors (e.g., voting), which goes beyond attracting, retaining, and engaging audience [20,27,36] or advertising [50,51]. With multiple audiences' hybrid consumption tendencies of consistency- and variety-seeking, our study advances the understanding of the trade-off between SMIs' reinforcement- and variation-seeking and identifies the sweet spot maximizing SMIs' influence over audiences in a planning horizon. As SMIs rising from grassroots differ in their capability and performance of video production, we find that low-performing SMIs benefit more from variation-seeking than from reinforcement-seeking. By contrast, the influence of high-performing SMIs on audiences could be maximized by either reinforcement- or variation-seeking, depending on the SMI type (herein SMI's signature content style). High-performing SMIs' variation-seeking (should it be taken) is encouraged earlier than later over a planning horizon. Our findings contribute to the research on content strategy by delving into the intertemporal process in which SMIs need exercise acute decision making based on ever-changing audience feedback and their own status on SVPs.

Second, our study advances the understanding about the heterogeneity of opposing content strategies in terms of SMI type and timing of decision making, which suggests possible observational learning [53] for different SMIs to identify the sweet spot of reinforcement-seeking and variation-seeking during a planning horizon. Recovering the dynamics between SMIs' content state transitions and their influence on audiences, our MDP model uncovers the heterogeneous effects of the two content strategies on SMIs' influence over audiences. With the parameter estimations, the MDP model produces time-varying optimal policies beyond the time horizon of the empirical observations (e.g., the 12-step optimal policies in Table 8 derived from the 5-step observations), which could predict SMIs' decisions at a future time to maximize their influence on audiences.

Apart from the decision making of content strategy, the MDP model

offers a computational approach to decision making that involves inter-temporal dynamics, such as strategically allocating resources to content production, dynamically pricing content on SVPs. Through modeling the context (state), process (action), and outcome (reward) of dynamic decision making, the MDP approach is capable of producing optimal strategy by leveraging empirical observations for parameter estimations (i.e., state transition and reward), instead of assuming that the parameters follow a specific distribution model such as probit, ordered logit, multinomial logistic, or Dirichlet distribution used in prior studies [20,69,70].

Third, this research adds to the currently scant literature on the new context of SVPs [1,45]. The access-based streaming media consumption embedded in social networks between SMIs and audiences creates an unprecedented digital space where SMIs are faced with both opportunities and challenges in producing content that expresses themselves, stands out from peers, and more importantly extends influence on audiences. The large-scale user engagement with streaming media renders this emerging context highly interactive and perpetually evolving. The diverse and networked audiences collapsed into one context complicate streaming media creators' forward-looking content production. As a call for more research attention to this niche but growing area, this paper delves into the context of streaming video to elucidate SMI's content strategies learned from the interactive environment.

Besides contributing to the academic literature, the findings of this paper also provide important practical implications to SMIs and SVPs. First, SMI's decision on reinforcement-seeking and variation-seeking needs to consider both enduring and time-varying factors, such as innate capabilities, ever-changing feedback, and opportunities lying ahead. Distinguishing low- and high-performing creators, our study suggests low-performing SMIs explore different content styles to maximize their influence on audiences. For certain types of SMIs who hardly benefit from variation-seeking, they need to be aware of the underlying risk of producing varied content.

Second, as SVPs embrace data analytics to improve decision making, the MDP model provides SVPs with a computational approach to capture inter-temporal dynamics between video content characteristics and content creators' influence on audiences platform-wise. Apart from audience's online votes for SMIs, our MDP approach, which is generalizable to other phenomena such as content sharing, content purchase, platform service subscription, could inform optimal content production

strategies for various objectives. The multidimensional features of video content combining both content and creator characteristics allow SVPs to analyze videos from viewer's subjective perceptions beyond objective acoustic features.

Third, given audiences' hybrid tendency to seek consistency and variety of media content, our findings have implications for SVP to make on-demand content recommendations for audiences. When audiences subscribe SMIs as loyal followers, SMIs' content strategies in terms of reinforcement-seeking and variation-seeking also affect followers' content consumption patterns, which could be considered by platform's recommendation algorithm. Instead of recommending content in accord with view history only, SVPs may vary recommended content based on audiences' dynamic preferences.

Our study has several limitations that nonetheless provide opportunities for future research. First, since our data set was collected from SMIs on a single SVP, studies using data sets from other platforms in the future could enhance the generalizability of our findings. Although the music videos analyzed in our study cover multiple languages (e.g., Chinese, English, and Russian) and genres that are universal across countries, the SMIs and audiences in our empirical setting exhibit the dynamics embedded in the creation and consumption of cultural content (i.e., musical performance), which bears both similarities and differences across countries and audience groups. Future research could further explore the heterogeneous content production strategies in different contexts. Second, this study has yet to take into account SMI's interactions with audiences via media channels besides the current streaming media environment. Future research can consider the spill-over effect of learned policies across multiple environments. Third, while the video/audio quality of musical performances is controlled by the producer of the competition program, it is worthwhile to recognize the importance of technical skills and knowledge (e.g., video/audio and camera-related techniques) on content quality, especially for grassroots who aim to make their way to SMI by producing videos on their own. The operationalizations and findings of our study are more applicable to content creators who are capable of maintaining the video/audio quality. With rapid advances in the equipment of video production, the technical barriers of video production are lowered so that SMI's success would rely more on strategic content than technical tools.

Notwithstanding the aforementioned limitations, our findings reveal that the tension between reinforcing the status quo and seeking a variation indeed exists in SMI's decision on streaming media content production. The optimal content strategies exhibit heterogeneity in SMI types and remaining time horizon. We hope that our paper paves the way for more research on the emerging and increasingly important digital environment in which access-based content consumption and streaming media-enabled user interactions call for SMI's strategic content production in an adaptive fashion.

CRedit authorship contribution statement

Ling Jiang: Conceptualization, Methodology, Data curation, Formal analysis, Software, Writing – original draft, Writing – review & editing. **Xingyu Chen:** Conceptualization, Writing – original draft, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Sentao Miao:** Conceptualization, Validation, Writing – original draft, Writing – review & editing. **Cong Shi:** Conceptualization, Methodology, Validation, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.dss.2023.114148>.

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